



NYU

Urban Heat & Cool Roofs in NYC

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PART 01

Project Overview

Rooftops & Urban Heat

Urban areas run hotter than their surroundings because dark impervious surfaces absorb and re-radiate solar heat instead of reflecting it. Rooftops alone account for 20–25% of urban surface area, making them one of the largest drivers of the Urban Heat Island effect (Akbari et al., 2001, Solar Energy).

The heat burden is unequal. Across 175 U.S. cities, people of color and low-income residents are consistently exposed to higher urban heat intensity than wealthier, whiter neighborhoods. In NYC, summer heat is the leading cause of weather-related death (Hsu et al., 2021, Nature Communications).

High-reflectivity roofs measurably reduce temperatures. NYC-specific modeling projects peak afternoon temperature reductions of up to 0.9°C under broad cool roof adoption. (Gaffin et al., 2012, Environmental Research Letters).

Rooftops & Urban Heat

NYC has operated a cool roof program since 2009, but there is no systematic, data-driven framework for identifying which buildings should be prioritized

Our analysis is designed to close that gap: characterizing the rooftop landscape, measuring whether installations have produced detectable temperature change, and developing an optimization framework for future targeting.

Research Questions & Approach

Methodology

Purpose

PCA & K-Means: Roof Typology Clustering

Do heat-vulnerable neighborhoods exhibit systematically lower rooftop reflectivity?

Time Series Analysis: Understanding surface temperature over time

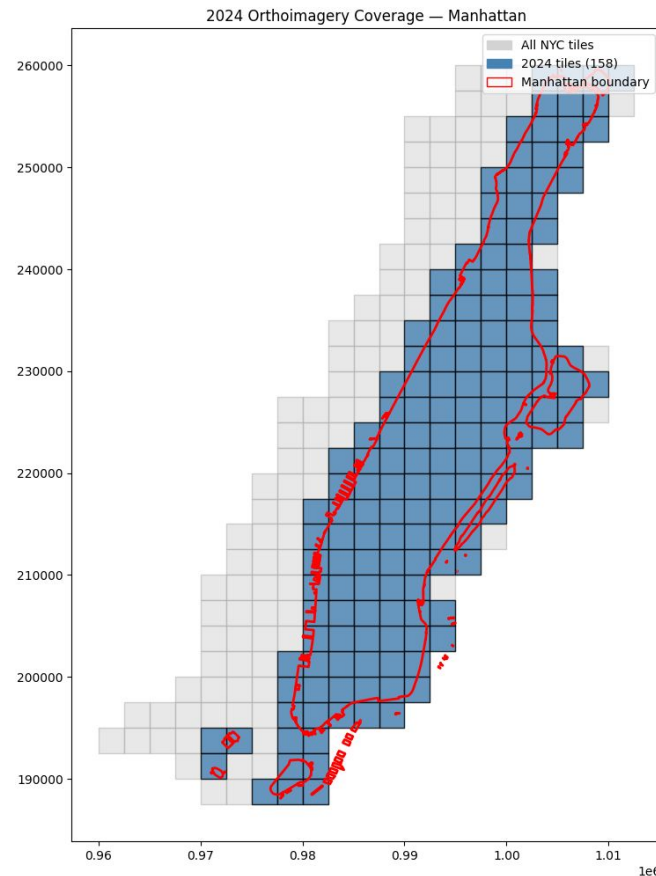
Have cool roof installations produced measurable reductions in land surface temperature?

Optimization: Targeting cool roof investment within budget

Which buildings represent the highest-impact targets for future installations under a fixed budget?

Data & Study Area

- Manhattan orthoimagery (2024)
 - 158 tiles, per building RGB extraction
- Primary Land Use Tax Lot Output (PLUTO) (2026)
 - Building footprint, land use data
- NYC CoolRoofs GIS Data
 - Participation records
- NASA MODIST LST
 - Land surface temperature, 2009-2024



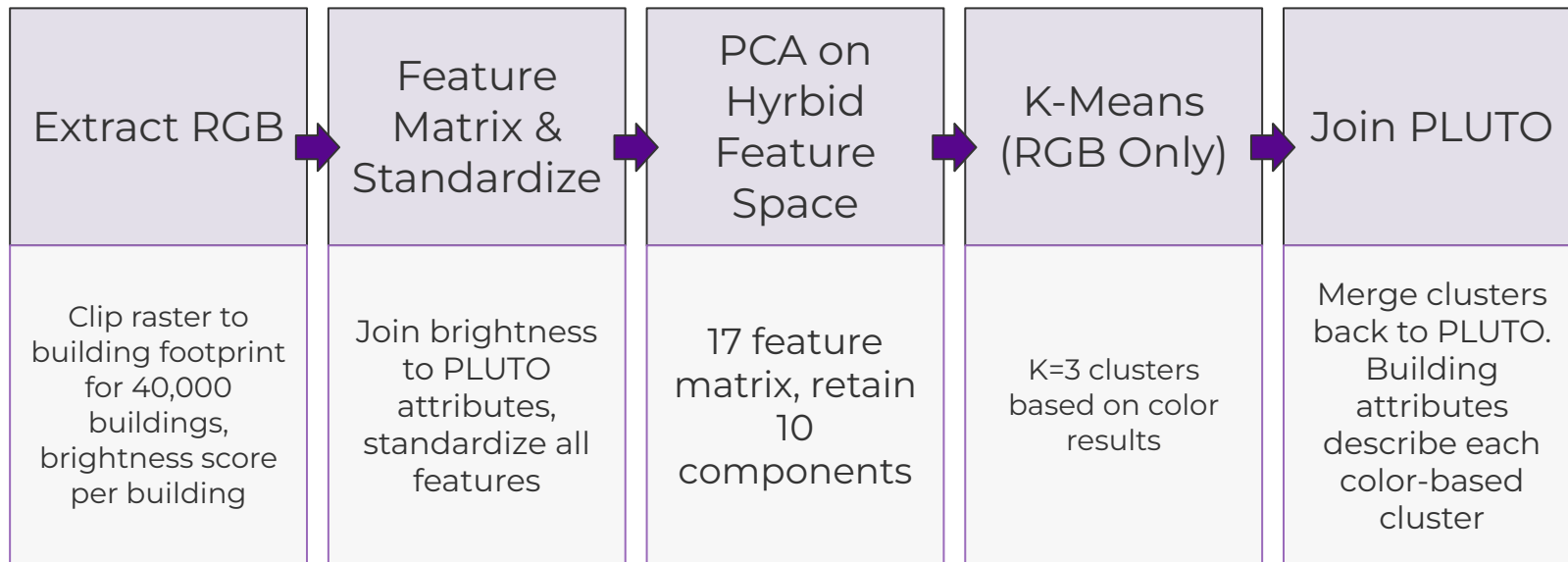
PART 02

Roof Typology Clustering

PCA & K-Means

*Compressing 18 building features into principal components,
and clustering buildings into distinct roof typology groups*

Methodology



Modeling Choices

01.

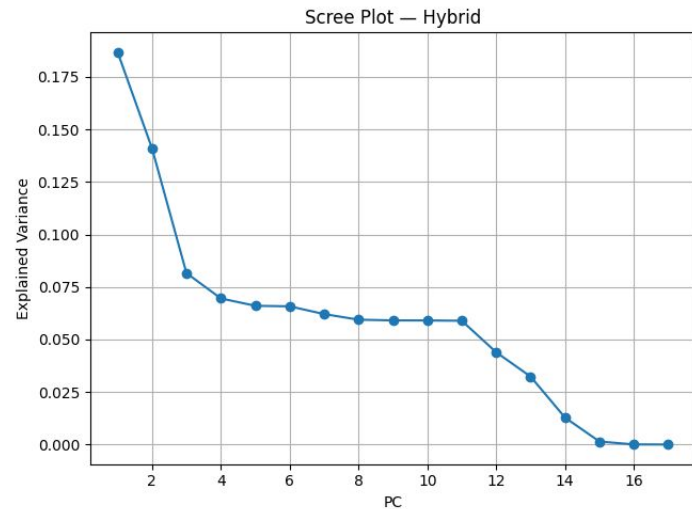
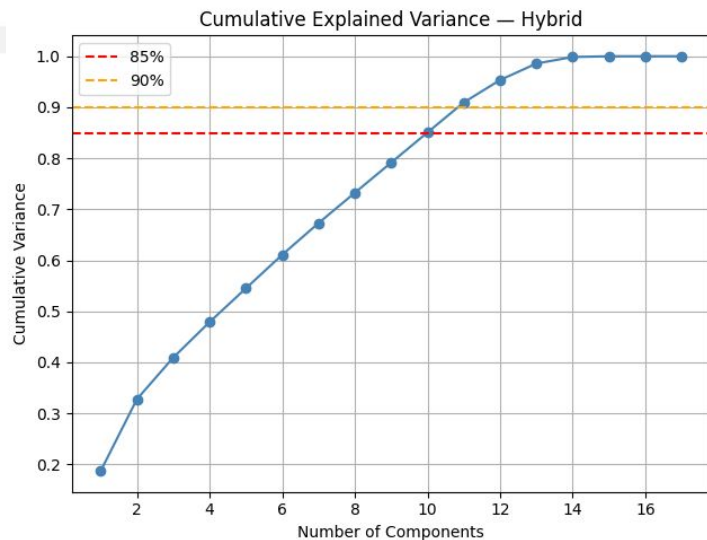
10 Components. Accounts for 85% of cumulative explained variance, and additional components capture noise and minor land use types.

02.

Cluster on RGB Only. Including building size conflated roof material with building type. RGB alone defines the optical typology. MapPLUTO then describes each cluster.

03.

K=3. There is no sharp elbow, so 3 clusters chosen for interpretation. Bright, medium, and dark clusters map well to different intervention priorities.



Results

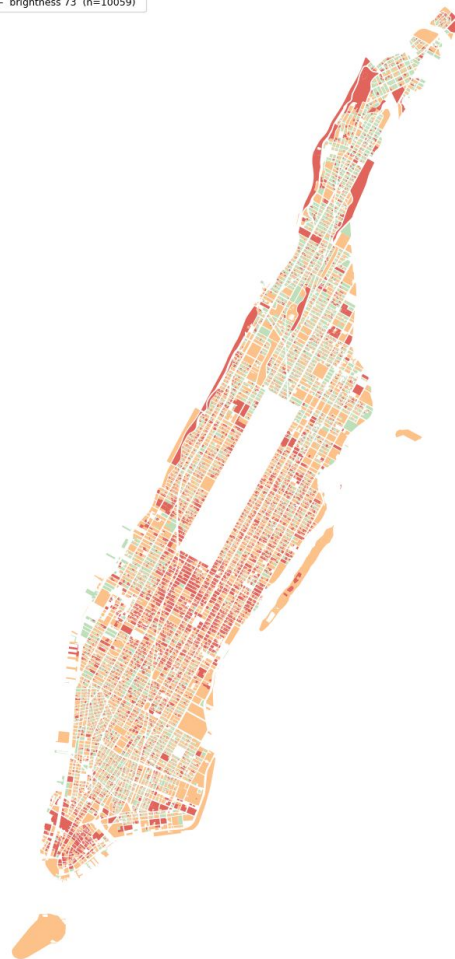
Brightest: Upper Manhattan

- Harlem South only 16% dark-roofed (mean brightness = 154)
- Older low-rise residential stock with naturally lighter roofing materials
- Washington Heights and East Harlem follow a similar pattern.
- These are all high heat-vulnerability neighborhoods

Darkest: Midtown & Lower Manhattan

- 72% of buildings dark-roofed (mean brightness = 73)
- Commercial land use dominates: large flat rooftop with highest solar heat load square foot

The darkest part of Manhattan is its commercial core not its most heat-vulnerable neighborhoods. This spatial mismatch between heat load and heat vulnerability is the tension the optimization addresses.

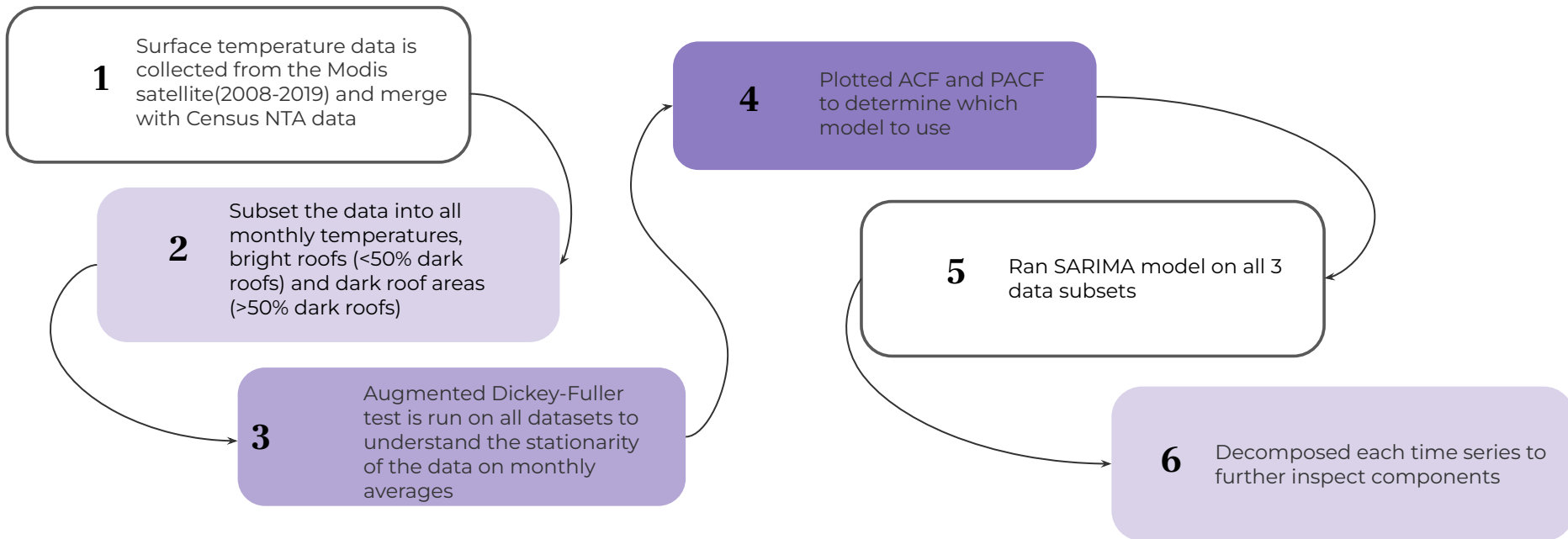


PART 03

Time-Series

Assessing cool roofs impact over time

Methodology

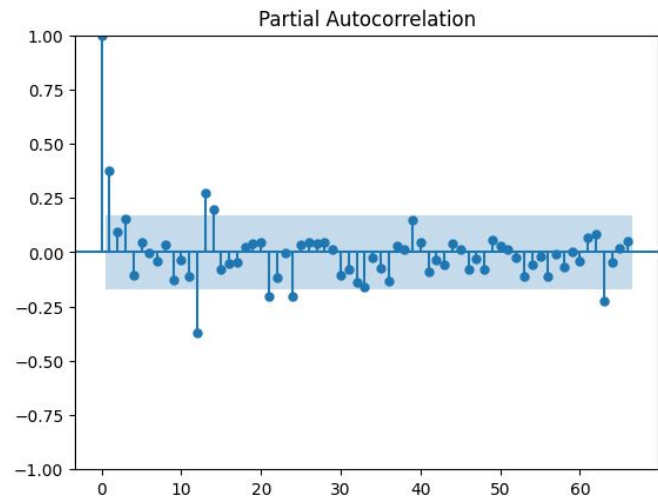
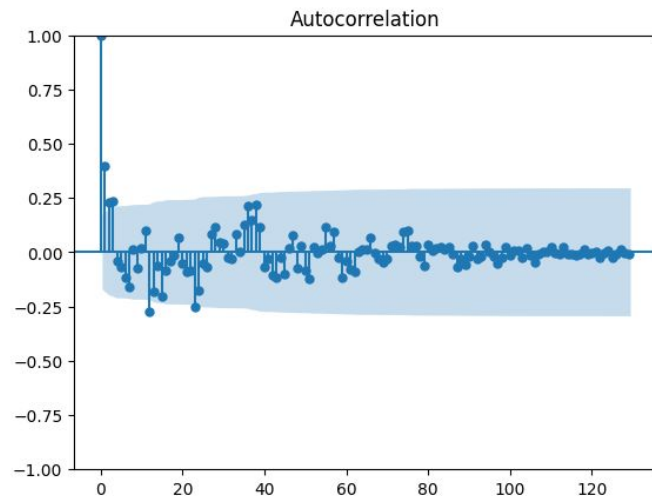


Results: ADF Test

Subset	ADF Statistic	P-Value
All NYC	-2.78	0.06
Bright Roofs	-2.77	0.06
Dark Roofs	-2.71	0.07

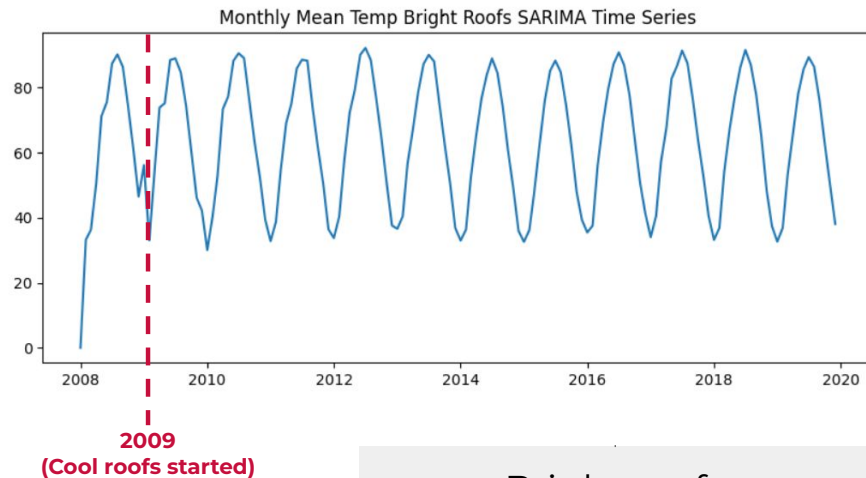
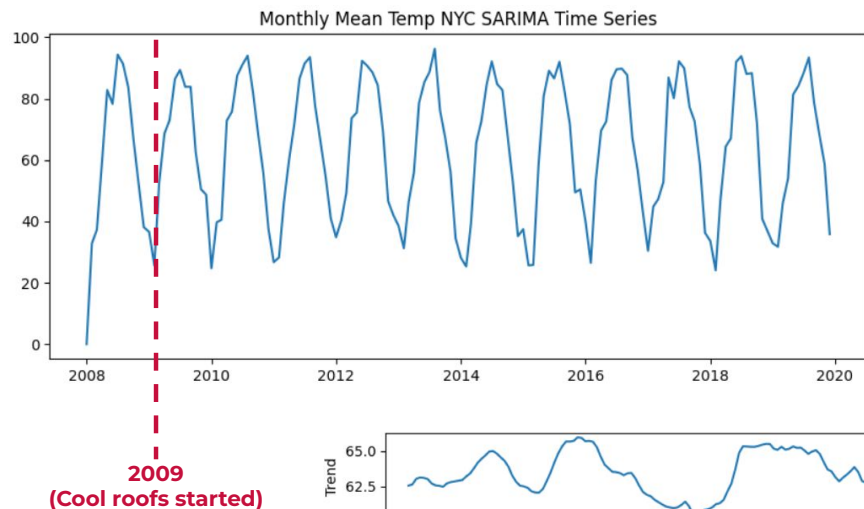
All three subset: overall, bright, and dark roofs, show the same result: **p-values are just above 0.05, confirming non-stationarity.** We differenced the data before fitting SARIMA.

Results: ACF & PACF (All Categories)



Some spikes (2010–2014) but none exceed significance threshold & this pattern held across all 3 subsets (bright, dark, all)

Results: SARIMA



While there is a clear link between Cool Roofs and localized cooling, the effect remains subtle and can be difficult to distinguish from other factors.

Limitation: MODSIS Satellite cannot gather information from smaller NTA's - This is a significant limitation of the study.

Interpretation

Cool roofs have not shifted NYC-wide surface temperatures at a city scale. Seasonal cycles dominate the signal and any post-2009 decline cannot be cleanly attributed to the program.

However, localized building-level cooling is real. AR memory was significant in the bright roof and all-NYC SARIMA models, but not in dark roof areas, suggesting bright roofs drive the temperature autocorrelation signal. **The program's impact shows up at the building scale, but in order to assess the city scale, more investigation is required.**

PART 04

Optimization

Cool Roofs installation plan

*Given a fixed \$24M budget,
which buildings should install cool roofs to maximize impact?*

Define the model

Decision Variables (Installed or not)

$$x_i \in \{0, 1\}$$

$$i = 1, \dots, N \text{ (} N = 40,013 \text{)}$$

0 = skip installation

1 = installation of cool roof

Cost Function (Installation Costs)

$$c_i = \bar{p} \cdot a_i$$

$\bar{p}$ = \$ 2.5/sqft. (midpoint of general
installation cost of \$1.5 - \$3.5)

a_i = roof area from NYC PLUTO
(sqft.)

Constraint (Budget)

$$C(\mathbf{x}) = \sum_{i=1}^N c_i \cdot x_i \leq \text{Budget}$$

Budget = \$24,000,000
(Actual budget in 2017)

Scenario A

Maximize Thermal Impact (Efficiency)

- Reduce the city's average temperature as much as possible, regardless of who lives in the building.
- Objective functions

$$\text{Maximize } Z_A = \sum \Delta T_i \cdot x_i$$

$$\Delta T_i = (\alpha_{\text{new}} - \alpha_i) \times 20^\circ\text{C}$$

$$x_i \in \{0, 1\}$$

$$\alpha_{\text{new}} = 0.8 \quad (\text{After cool roof: } \sim 15^\circ\text{C})$$

$$\alpha_i = \text{current albedo assumed by RGB}$$

$$= 35 + (0.30 - \text{Albedo}) \times 40$$

$$* \text{ Albedo} = \frac{\text{Brightness}}{255}$$

$$0 = \text{skip installation}$$

$$1 = \text{installation of cool roof}$$

* Assumed a linear relationship where one albedo shift would lower the surface temperature by 20°C

Scenario B

Minimize HVI (Heat Vulnerability Index)

- Prioritize cooling in neighborhoods with the highest HVI (Heat Vulnerability Index) scores.
- Objective functions

$$\text{Minimize } Z_B = \sum \Delta w_i \cdot x_i$$

$$\Delta w_i \in \{1, 2, 3, 4, 5\}$$

$$x_i \in \{0, 1\}$$

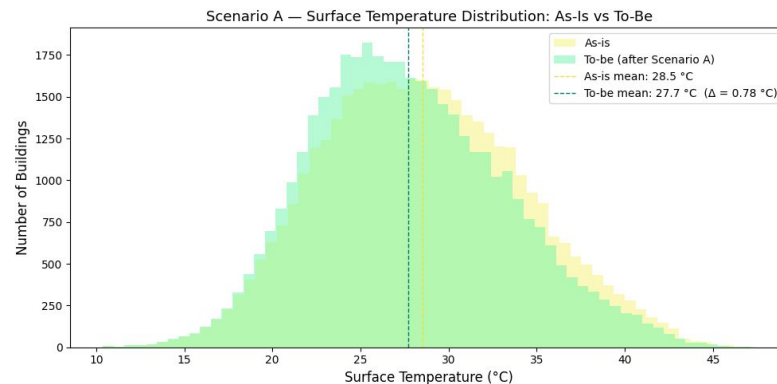
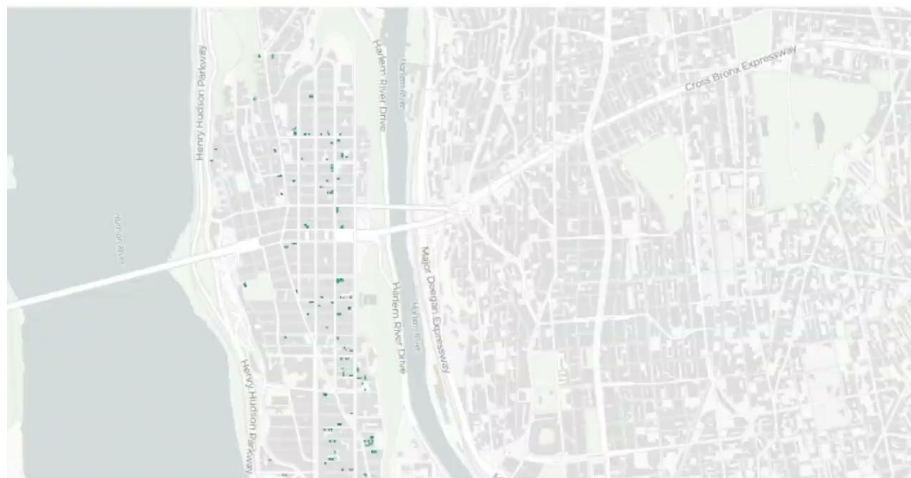
$$* \text{ HVI index}$$

$$0 = \text{skip installation}$$

$$= 1 \text{ (lowest risk)} \sim 5 \text{ (highest risk)} \quad 1 = \text{installation of cool roof}$$

Scenario A: Results

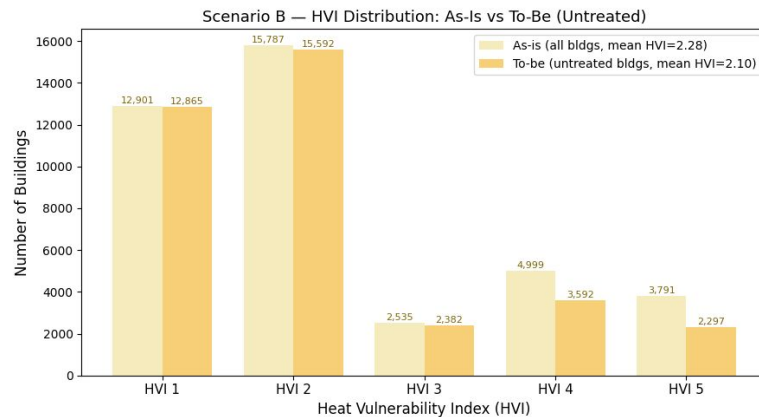
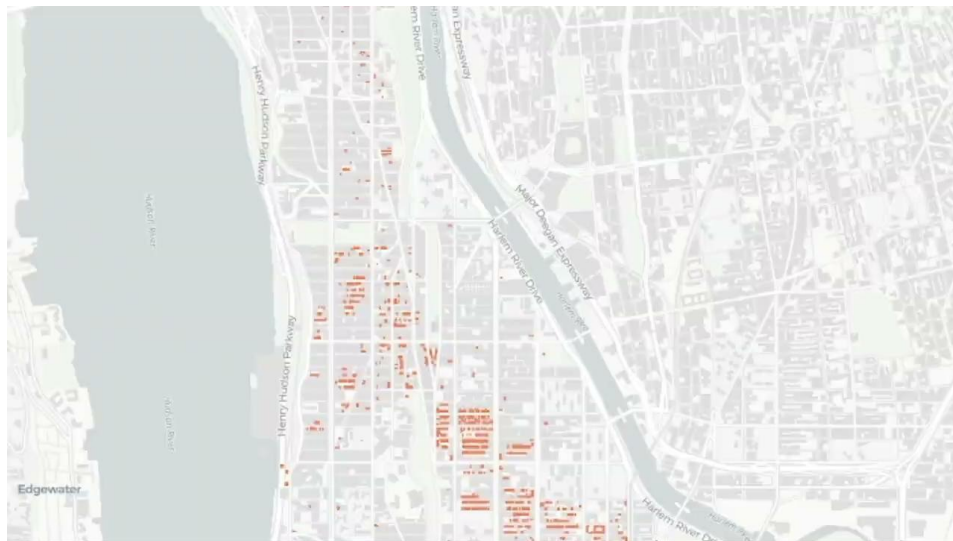
Scenario A: improve temperature



*** 3,370 buildings selected**

	Mean (°C)	Total (°C)
As-is	28.50	1,139,642.3
To-be	27.72	1,108,252.4
Reduction	-0.78	31,389.9

Scenario B: Results



Mean (HVI)

As-is 2.275

To-be 2.098

Reduction -0.177

*** 3,285 buildings selected**
HVI improvement : 14,000

Comparative analysis

Cool Roof Selection Analysis: Scenario A (temperature) vs Scenario B (HVI)



Scenario A : 3,370 buildings selected

Scenario B : 3,285 buildings selected

- **Overlap:** 1,263 buildings
- **Only in A:** 2,107 buildings
- **Only in B:** 2,022 buildings

Jaccard similarity: 23.4%

** the intersection size divided by the union size,
also called intersection over union (IoU)*

PART 05

Conclusion

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From Passive to Proactive

Currently, the NYC Cool Roofs program relies on voluntary participation with no targeting.

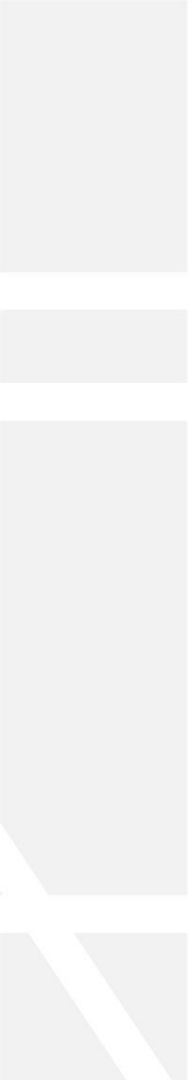
Our optimization analysis shows that **a \$24M budget, directed by PCA-identified roof typologies and heat vulnerability, could cool nearly 3,400 buildings**, far more effectively than random selection.

Toward a Resilient NYC

Building-scale cooling is real and measurable.

To strengthen the program, future work should extend the time series beyond 2019, incorporate finer-resolution temperature data, and **use our priority framework to target installations in the highest-risk neighborhoods first** - moving NYC toward a more resilient future.

Limitations: Cool roof impacts are difficult to isolate at city scale given coarse satellite resolution, approximate installation dates, and confounding factors like weather variability and urban development. Future work with finer-resolution data and longer time series would strengthen causal attribution.



Appendix

**Data Sources, Literature Review,
Additional Figures**

Data Sources

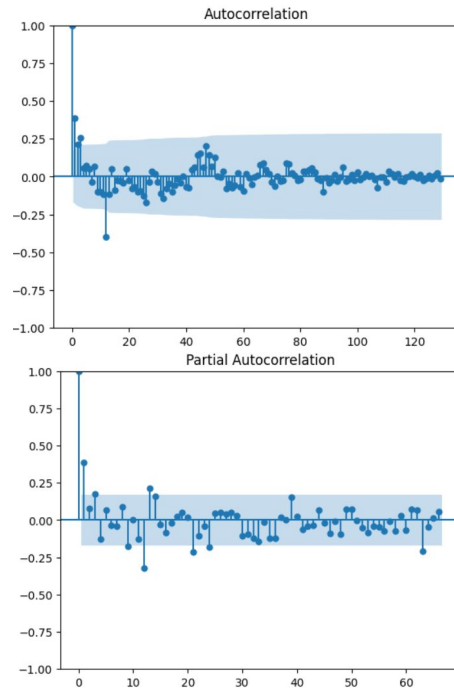
- Manhattan Orthoimagery (2024) : High-resolution aerial imagery consisting of 158 tiles used for per-building RGB color extraction to determine rooftop reflectivity.
- Primary Land Use Tax Lot Output (PLUTO) (2026) : Used for building footprints, land use data, and building-level characteristics such as age and size.
- Census Neighborhood Tabulation Area (NTA) Data : Demographic data including income and race, merged with shapefiles (nybb and NTA) to conduct neighborhood-level analysis.
- Heat Vulnerability Index (HVI) : Scores used to prioritize cooling interventions in neighborhoods at the highest risk for heat-related illness and death.

Literature Review

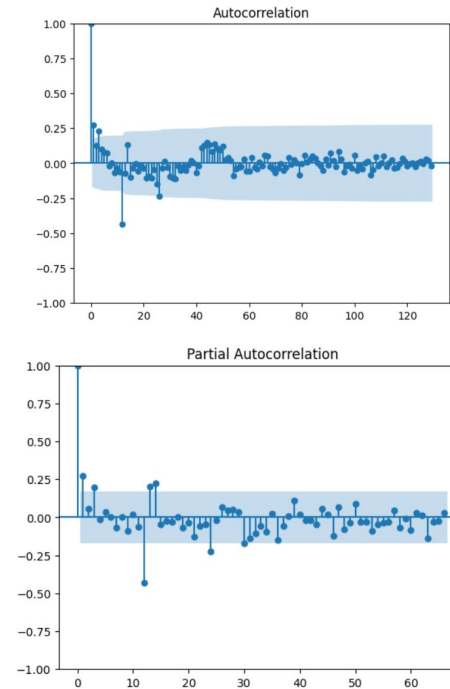
- Akbari et al. (2001), Solar Energy : Found that rooftops account for 20–25% of urban surface area, establishing them as a primary driver of the Urban Heat Island (UHI) effect.
- Hsu et al. (2021), Nature Communications : Investigated the unequal heat burden across 175 U.S. cities, noting that summer heat is the leading cause of weather-related death in NYC, particularly affecting low-income residents and people of color.
- Gaffin et al. (2012), Environmental Research Letters : Provided NYC-specific modeling that projected peak afternoon temperature reductions of up to 0.9°C through broad adoption of cool roofs.

1. Time Series Figures: ACF & Partial ACF

A. Bright Category

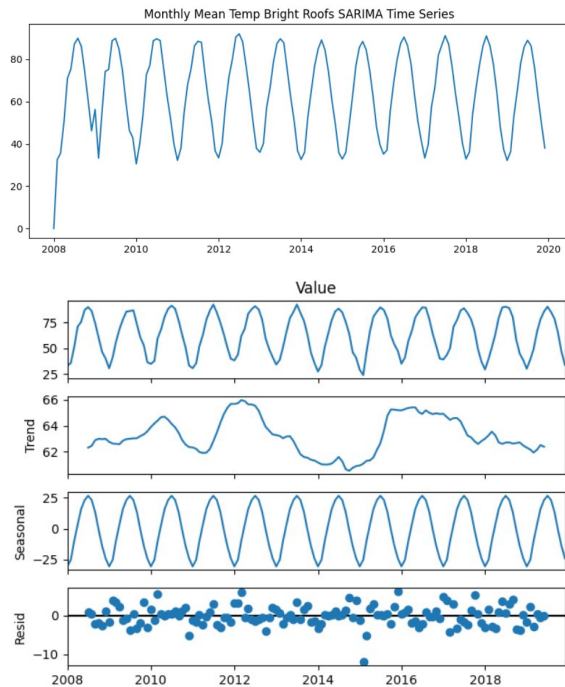


B. Dark Category

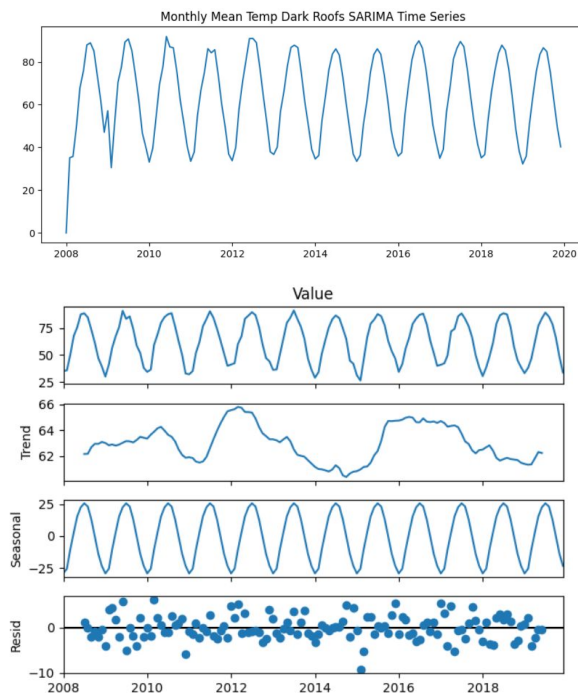


2. Time Series Figures for SARIMA, Trend, Seasonal and Residual

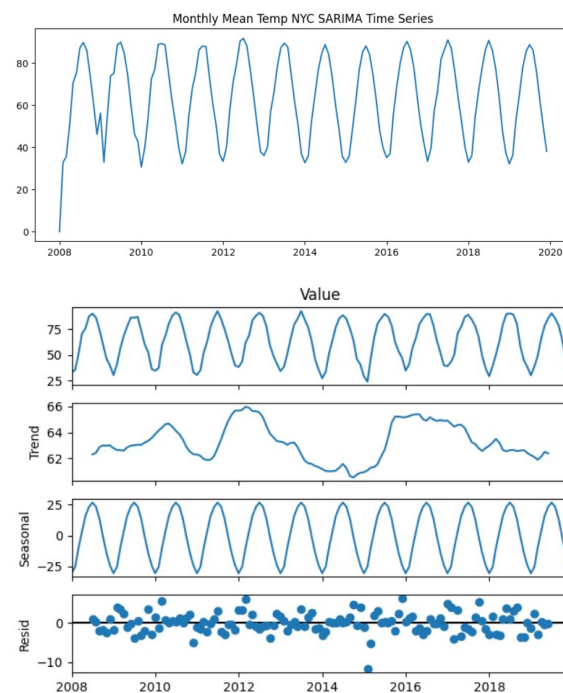
A. Bright Category



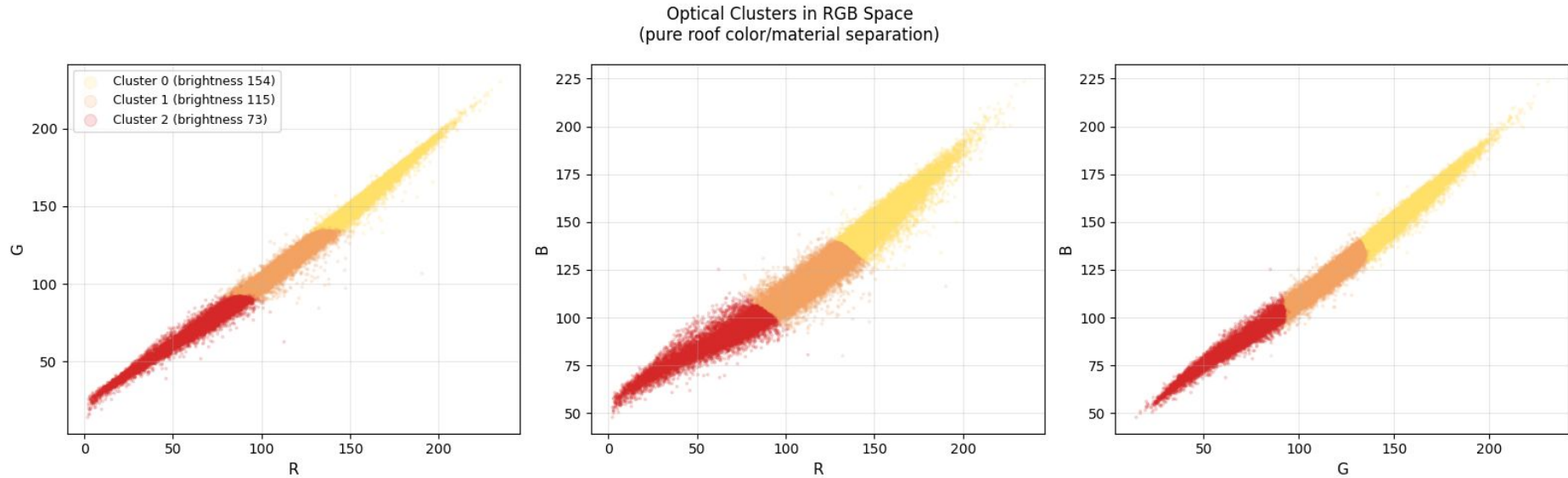
B. Dark Category



C. Overall

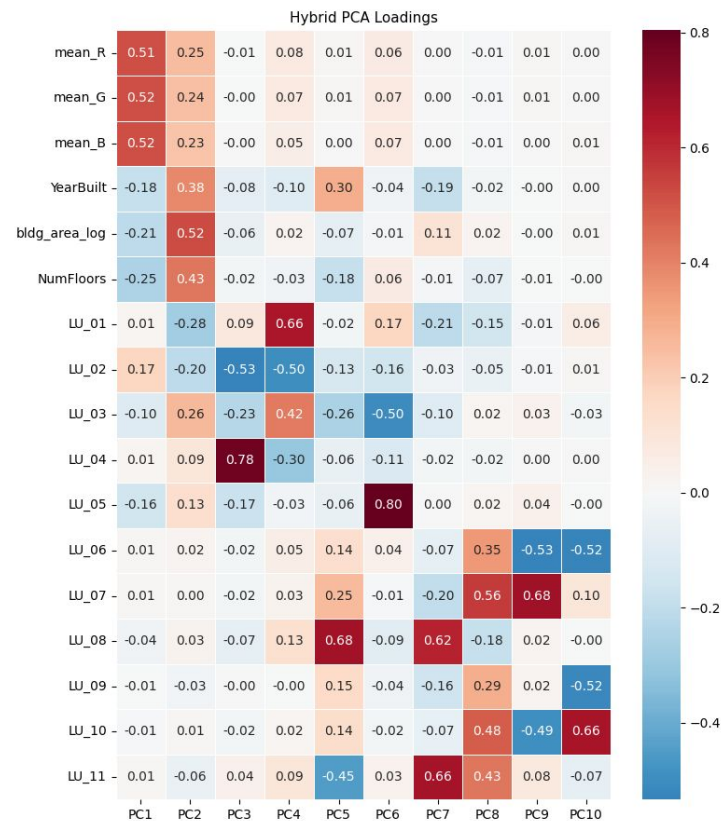
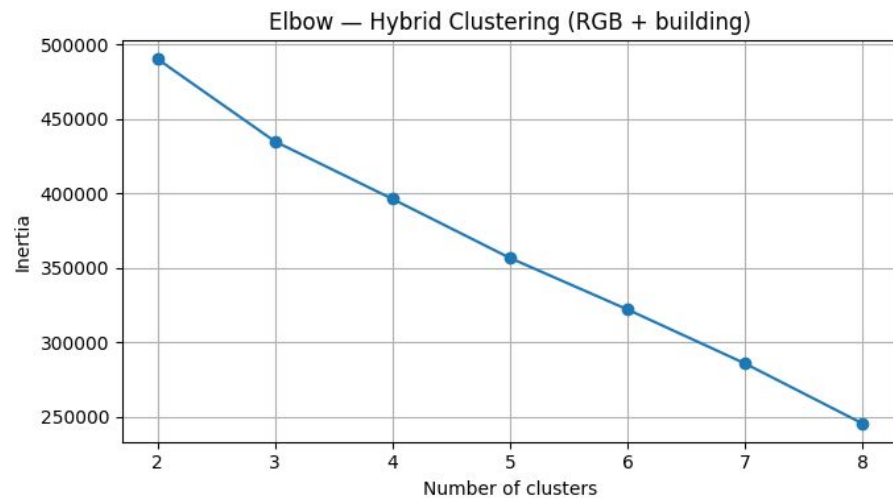


3. PCA Figures for Optical Clusters in 3D Space

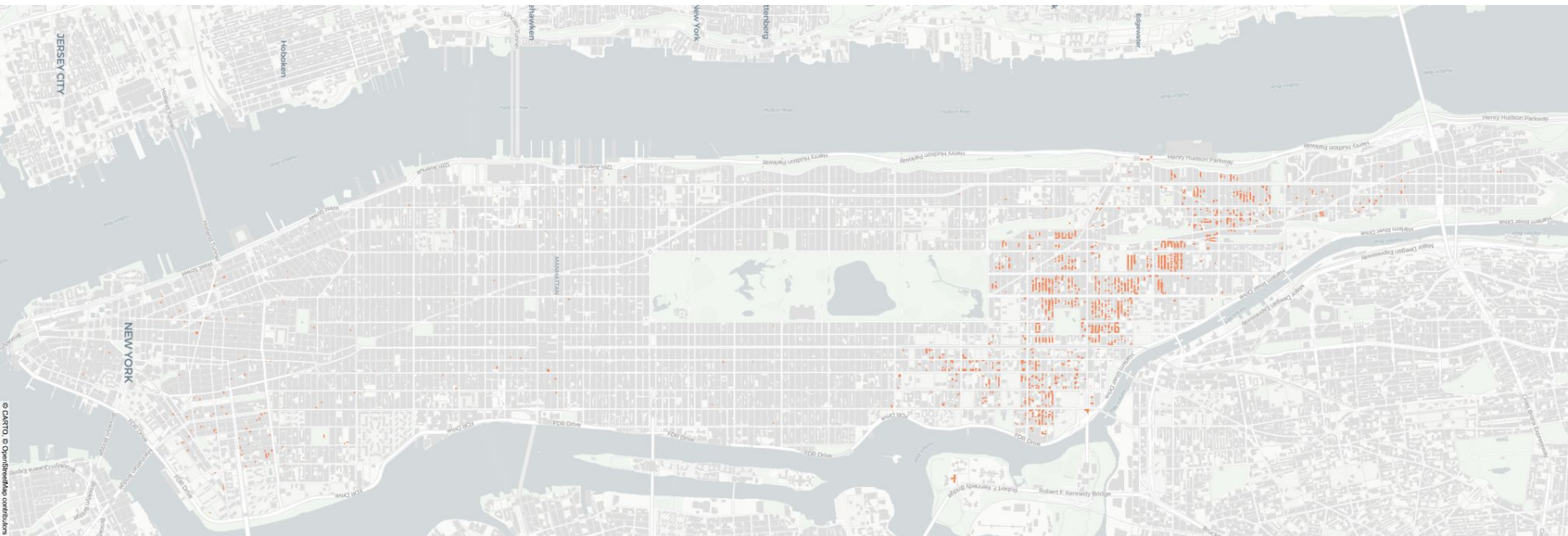


Clusters plot in 3D space to confirm they are spatially separated by roof color, not by building size

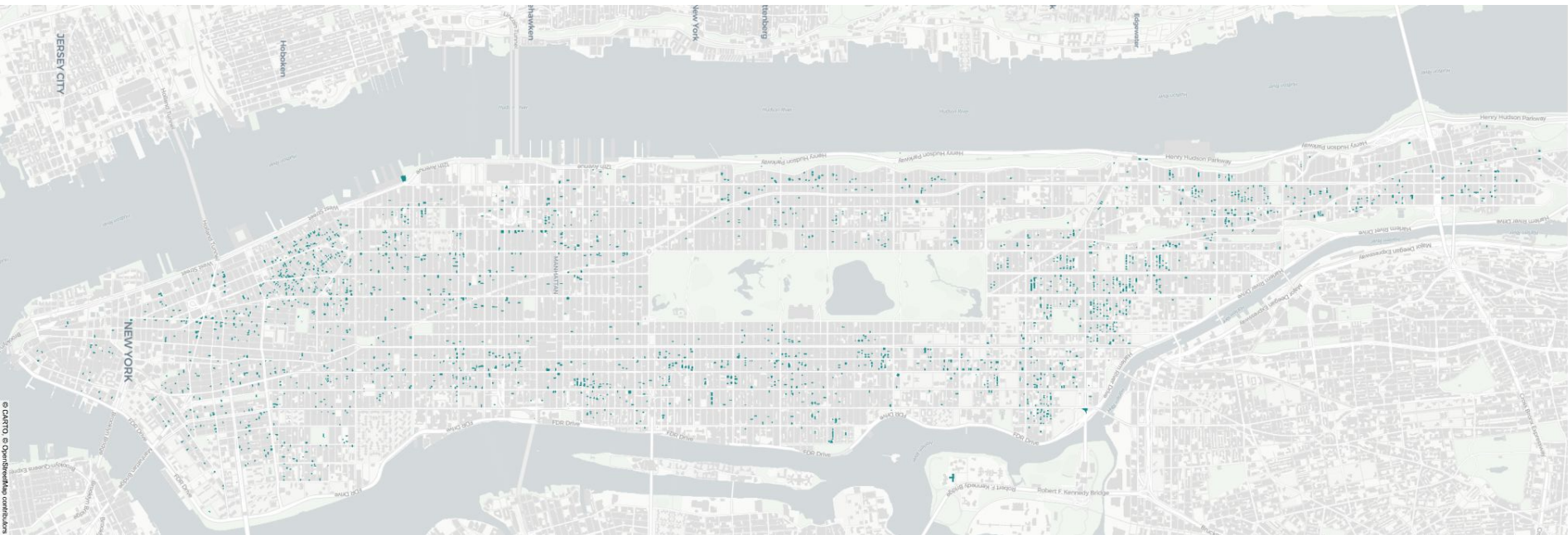
4. PCA Figures for Hybrid PCA Heatmap & Elbow Plot



5. Optimization Figures: Final map from scenario A



6. Optimization Figures: Final map from scenario B



7. Optimization Figures: Final map from overlapping two scenarios

